

## LOPAMUDRA GIRI

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H-index-18, i-10 index-27, Total number of citations-1164 (Google Scholar)

### Research Interests

**Live Cell Imaging:** Live cell imaging using laser scanning confocal microscopy, 3D imaging, Imaging of protein aggregation, Calcium imaging, non-animal model (NAMs) development, studying neurodegeneration using hypoxic model.

**Systems Biology:** Modeling of calcium signaling, Biochemical oscillations, parameter estimation for oscillatory system, Genetic algorithm.

**AI/ML in quantitative microscopy:** Clustering and Classification of time series data, Machine learning and Deep learning (convolution neural network) for automation in image analysis.

**Fabrication of imaging device using 3D Printing:** Fabrication of milifluidic device for perfusion culture, coculture, imaging device for detection of protein from serum (Using stereolithography/fused deposition modeling-based template).

### Professional Experience

| S.No. | Position             | Institution                                    | Field                | Experience                                                                                                                 | Year             |
|-------|----------------------|------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------|------------------|
| 1     | Associate Professor  | Indian Institute of Technology                 | Chemical Engineering | Imaging of condensates in neurons/axons, Age-related disease, Systems Biology, Mili fluidic device, automation using ML/DL | 2019 Jan-current |
|       | Assistant Professor  | Indian Institute of Technology, Hyderabad      | Chemical Engineering | Bioimaging (LSCM), Systems biology, Machine learning.                                                                      | 2013-2018        |
|       | Post-doctoral Fellow | Washington University, St. Louis, USA          | Anesthesiology       | GPCR targeting pathway, confocal microscopy                                                                                | 2011-2012        |
| 2     | Post-doctoral Fellow | University of Illinois, Urbana, Champaign, USA | Microbiology         | Drug screening against neurotoxin using FRET.                                                                              | 2009-2010        |

### Educational Qualification

| S.No. | Degree           | Institution, Location                                 | Field                                | Year      | Remarks |
|-------|------------------|-------------------------------------------------------|--------------------------------------|-----------|---------|
| 1.    | PhD              | University of Iowa, Iowa city, IA, USA                | Chemical and Biochemical Engineering | 2004-2009 | 4.0/4.0 |
| 2.    | M.Tech           | Indian Institute of Technology, Bombay, Mumbai, India | Chemical Engineering                 | 2002-2004 | 9.7/10  |
| 3.    | B.Tech. (Honors) | Jadavpur University, Calcutta, India                  | Chemical Engineering                 | 1997-2001 | 3.7/4.0 |

### Awards/honors and Fellowships

1. Royal Academy of Engineering, UK, Transforming Systems through Partnership 23/25 (India) Contract, **(2023-2026)**, 108,000 SP.
2. Accredited supervisor at Swinburne University of Technology, Melbourne, Australia, **(2020)**.
3. Recognition in honor for remarkable contribution as an author to **IEEE** publication, IEEE, Advancing Technology for Humanity, **(2025)**.
4. Karl Kammermeyer Graduate Award for excellence in research and scholarship, Univ. of Iowa, **(2009)**.
5. Center for Bioprocessing and Biocatalysis (CBB) fellowship, University of Iowa, **(2006 – 2009)**.

#### Student Awards: (2020-2025)

1. Research Excellence Award: Vaibhav Dhyani, PhD student, **(2024)**.
2. Conference Support for exceptional research: Debasmita Saha, **(2022)**.
3. Drishti CPS Fellowship, IIT, Indore, Sayantan Ghosh **(2023-25)**.
4. Bio nest Hackathon, First Prize, IIT Guwahati, Sayantan Ghosh, Guwahati **(2022)**.
5. Best Paper Presentation in All India Ophthalmological Conference, Sayantan Ghosh, **(2022)**.

## Industry experiences

| Duration        | Institution                                                     | Specialization and expertise                                                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2007<br>May-Aug | Summer Intern, <i>Eli Lilly and Company</i> , Indianapolis, USA | Tailored Therapy: High throughput assay in multi-well plate for “ <i>in vitro</i> drug metabolism” using Lilly drug compounds, Characterization of the impact of genetic variants of CYP enzymes on drug metabolism.                         |
| 2002            | <i>Cognizant Technology Solutions</i> , Calcutta, India         | Programmer Analyst, Designed modules for customer relationship management software for JPMorgan Chase. <i>Training</i> : Algorithms, C programming, database, Data structure, SQL, quality control, software engineering and project design. |

## Publications (Journal)

### Section A: List of publications before joining IITH

2004- 2013

1. Karunarathne, W.A., **Giri, L.**, Patel, A.K., Venkatesh, K.V. and Gautam, N., 2013. Optical control demonstrates switch-like PIP3 dynamics underlying the initiation of immune cell migration. *Proceedings of the National Academy of Sciences*, 110(17), pp. E1575-E1583. **Impact factor: 9.1 (Q1)**
2. O'Neill, P.R., **Giri, L.**, Karunarathne, W.A., Patel, A.K., Venkatesh, K.V. and Gautam, N., 2014. The structure of dynamic GPCR signaling networks. *Wiley Interdisciplinary Reviews: Systems Biology and Medicine*, 6(1), pp.115-123. **Impact factor: 6.98 (Q1)**
3. WKarunarathne, W.A., **Giri, L.**, Kalyanaraman, V. and Gautam, N., 2013. Optically triggering spatiotemporally confined GPCR activity in a cell and programming neurite initiation and extension. *Proceedings of the National Academy of Sciences*, 110(17), pp.E1565-E1574. **Impact factor: 9.1 (Q1)**
4. **Giri, L.**, Feiss, M.G., Bonning, B.C. and Murhammer, D.W., 2012. Production of baculovirus defective interfering particles during serial passage is delayed by removing transposon target sites in fp25k. *Journal of general virology*, 93(2), pp.389-399. **Impact factor: 4.3 (Q2)**
5. **Giri, L.**, Li, H., Sandgren, D., Feiss, M.G., Roller, R., Bonning, B.C. and Murhammer, D.W., 2010. Removal of transposon target sites from the Autographa californica multiple nucleopolyhedrovirus fp25k gene delays, but does not prevent, accumulation of the few polyhedra phenotype. *Journal of general virology*, 91(12), pp.3053-3064. **Impact factor: 4.3 (Q2)**
6. **Giri, L.**, Mutalik, V.K. and Venkatesh, K.V., 2004. A steady state analysis indicates that negative feedback regulation of PTP1B by Akt elicits bistability in insulin-stimulated GLUT4 translocation. *Theoretical Biology and Medical Modelling*, 1(1), p.2. **Impact factor: 2.4 (Q2)**

### Section B: List of publications after becoming Assistant Professor with students from IITH as co-authors.

2014

7. **Giri, L.**, Patel, A.K., Karunarathne, W.A., Kalyanaraman, V., Venkatesh, K.V. and Gautam, N., 2014. A G-protein subunit translocation embedded network motif underlies GPCR regulation of calcium oscillations. *Biophysical journal*, 107(1), pp.242-254. **Impact factor: 3.1 (Q1)**

2017

8. Gupta, R.K., Swain, S., Kankanamge, D., Priyanka, P.D., Singh, R., Mitra, K., Karunarathne, A. and **Giri, L.**, 2017. Comparison of calcium dynamics and specific features for G protein-coupled receptor-targeting drugs using live cell imaging and automated analysis. *SLAS DISCOVERY: Advancing Life Sciences R&D*, 22(7), pp.848-858, **Impact factor: 2.7 (Q1)**

9. Rathi, S., Jalali, S., Patnaik, S., Shahulhameed, S., Musada, G.R., Balakrishnan, D., Rani, P.K., Kekunnaya, R., Chhablani, P.P., Swain, S. and **Giri, L.**, 2017. Abnormal complement activation and inflammation in the pathogenesis of retinopathy of prematurity. *Frontiers in immunology*, 8, p.1868. **Impact factor: 5.9 (Q1)**
10. Singh, R., Miriyala, S.S., **Giri, L.**, Mitra, K. and Kareenhalli, V.V., 2017. Identification of unstructured model for subtilin production through *Bacillus subtilis* using hybrid genetic algorithm. *Process Biochemistry*, 60, pp.1-12. **Impact factor: 4.0 (Q2)**

## 2018

11. Swain, S., Gupta, R.K., Ratnayake, K., Priyanka, P.D., Singh, R., Jana, S., Mitra, K., Karunarathne, A. and **Giri, L.**, 2018. Confocal imaging and k-means clustering of GABAB and mGluR mediated modulation of Ca<sup>2+</sup> spiking in hippocampal neurons. *ACS chemical neuroscience*, 9(12), pp.3094-3107. **Impact factor 3.9 (Q1)**
12. Saxena, A., Byram, P.K., Singh, S.K., Chakraborty, J., Murhammer, D. and **Giri, L.**, 2018. A structured review of baculovirus infection process: Integration of mathematical models and biomolecular information on cell-virus interaction. *Journal of General Virology*, 99(9), pp.1151-1171. **Impact factor: 4.3 (Q2)**

## Section B: List of publications after becoming Associate Professor with students from IITH as co-authors

## 2019

13. Shahulhameed, S., Swain, S., Jana, S., Chhablani, J., Ali, M.J., Pappuru, R.R., Tyagi, M., Vishwakarma, S., Sailaja, N., Chakrabarti, S. and **Giri, L.**, 2020. A robust model system for retinal hypoxia: live imaging of calcium dynamics and gene expression studies in primary human mixed retinal culture. *Frontiers in Neuroscience*, 13, p.1445. **Impact factor 3.2 (Q2)**

## 2020

14. Venkateswarlu, K., Suman, G., Dhyani, V., Swain, S., **Giri, L.** and Samavedi, S., 2020. Three-dimensional imaging and quantification of real-time cytosolic calcium oscillations in microglial cells cultured on electrospun matrices using laser scanning confocal microscopy. *Biotechnology and Bioengineering*, 117(10), pp.3108-3123. **Impact factor 3.6 (Q2)**
15. Jain S, Swain S, Das L, Swain S, **Giri L**, Kondapi AK, Narayanan Unni H, Microfluidic Protein Imaging Platform: Study of Tau Protein Aggregation and Alzheimer Drug Response. (2020) *Bioengineering* 7(4):162. **Impact factor 3.7(Q2)**
16. Singh, R. Sharma, S.R., K., Kareenhali, V. V, Mitra, K., **Giri, L.**, Mitra., Experimental investigation into indole production using passaging of *E. coli* and *B. subtilis* along with unstructured modeling and parameter estimation using dynamic optimization: An integrated framework, 2020, *Biochemical Engineering Journal*, 163, **Impact factor:3.8 (Q2)**
17. Dhyani, V., Gare, S., Gupta, R.K., Swain, S., Venkatesh, K.V. and **Giri, L.**, 2020. GPCR mediated control of calcium dynamics: A systems perspective. *Cellular signalling*, 74, p.109717. **Impact factor 3.7 (Q2)**
18. Saxena, A., Ravutla, S., Upadhyay, V., Jana, S., Murhammer, D. and **Giri, L.**, 2020. Statistical modeling of cell-to-cell variability in viral infection during passaging in suspension cell culture: Application in Monte-Carlo simulation. *Biotechnology and bioengineering*, 117(5), pp.1483-1501. **Impact factor: 3.6 (Q2)**
19. Das, A., Dutta, S., Sen, M., Saxena, A., Kumar, J., **Giri, L.**, Murhammer, D.W. and Chakraborty, J., 2021. A detailed model and Monte Carlo simulation for predicting DIP genome length distribution in baculovirus infection of insect cells. *Biotechnology and bioengineering*, 118(1), pp.238-252. **Impact factor: 3.6 (Q2)**

## 2021

20. Manohar, K., Gupta, R.K., Gupta, P., Saha, D., Gare, S., Sarkar, R., Misra, A. and **Giri, L.**, 2021. FDA approved L-type channel blocker Nifedipine reduces cell death in hypoxic A549 cells through modulation of mitochondrial calcium and superoxide generation. *Free Radical Biology and Medicine*, 177, pp.189-200. **Impact factor: 8.2 (Q1)**
21. Kankanamge, D., Ubeyasinghe, S., Tennakoon, M., Pantula, P.D., Mitra, K., **Giri, L.** and Karunarathne, A., 2021. Dissociation of the G protein  $\beta\gamma$  from the Gq-PLC $\beta$  complex partially attenuates PIP2 hydrolysis. *Journal of Biological Chemistry*, 296. **Impact factor: 3.9 (Q1)**
22. Manohar, K., Gare, S., Chel, S., Dhyani, V. and **Giri, L.**, 2021. Quantitative confocal microscopy for grouping of dose-response data: deciphering calcium sequestration and subsequent cell death in the presence of excess norepinephrine. *SLAS TECHNOLOGY: Translating Life Sciences Innovation*, 26(5), pp.454-467. **Impact factor: 3.7 (Q2)**

## 2022

23. Gare, S., Chel, S., Abhinav, T.K., Dhyani, V., Jana, S. and **Giri, L.**, 2022. Mapping of structural arrangement of cells and collective calcium transients: an integrated framework combining live cell imaging using confocal microscopy and UMAP-assisted HDBSCAN-based approach. *Integrative Biology*, 14(8-12), pp.184-203. **Impact factor: 2.9 (Q3)**
24. Das, S., Saha, D., Majumdar, S. and **Giri, L.**, 2022. Imaging methods for the assessment of a complex hydrogel as an ocular drug delivery system for glaucoma treatment: Opportunities and challenges in preclinical evaluation. *Molecular Pharmaceutics*, 19(3), pp.733-748. **Impact factor: 4.5 (Q1)**
25. Dey, A., Gare, S., Swain, S., Bhattacharya, P., Dhyani, V., **Giri, L.** and Neogi, S., 2022. 3D imaging and quantification of PLL coated fluorescent ZnO NP distribution and ROS accumulation using laser scanning confocal microscopy. *AIChE Journal*, 68(9), p.e17801. **Impact factor: 4.0 (Q1)**

## 2023

26. Sharma, S., Mahadevan, J., **Giri, L.** and Mitra, K., 2023. Identification of optimal flow rate for culture media, cell density, and oxygen toward maximization of virus production in a fed-batch baculovirus-insect cell system. *Biotechnology and Bioengineering*, 120(12), pp.3529-3542. **Impact factor: 3.6 (Q2)**
27. Dhyani, V., Kumar, S., Manne, S.R., Kaur, I., Jana, S., Russell, S., Sarkar, R. and **Giri, L.**, 2023. Three-Dimensional Tracking of Intracellular Calcium and Redox State during Real-Time Control in a Hypoxic Gradient in Microglia Culture: Comparison of the Channel Blocker and Reoxygenation under Ischemic Shock. *ACS chemical neuroscience*, 14(10), pp.1810-1825. **Impact factor: 3.9 (Q1)**
28. Dhyani, V., George, K., Gare, S., Venkatesh, K.V., Mitra, K. and **Giri, L.**, 2020. A computational model to uncover the biophysical underpinnings of neural firing heterogeneity in dissociated hippocampal cultures. *Hippocampus*, 2023. **Impact factor: 2.7 (Q1)**
29. Saha, D., Vishwakarma, S., Gupta, R.K., Pant, A., Dhyani, V., Sharma, S., Majumdar, S., Kaur, I. and **Giri, L.**, 2023. Non-prophylactic resveratrol-mediated protection of neurite integrity under chronic hypoxia is associated with reduction of Cav1.2 channel expression and calcium overloading. *Neurochemistry International*, 164, p.105466. **Impact factor: 4.0 (Q2)**
30. Sharma, S., Sarkar, R., Mitra, K. and **Giri, L.**, 2023. Computational framework to understand the clinical stages of COVID-19 and visualization of time course for various treatment strategies. *Biotechnology and bioengineering*, 120(6), pp.1640-1656. **Impact factor: 3.6 (Q2)**
31. Das, S., **Giri, L.** and Majumdar, S., 2023. Hofmeister series: An insight into its application on gelatin and alginate-based dual-drug biomaterial design. *European Polymer Journal*, 189, p.111961. **Impact factor: 6.3 (Q1)**

## 2024

32. Ande, S., Avasarala, S., Swain, S., Karunarathne, A., **Giri, L.** and Jana, S., 2024. Robust entropy rate estimation for nonstationary neuronal calcium spike trains based on empirical probabilities. *Journal of Neural Engineering*, 21(5), p.056038. **Impact factor 3.8 (Q1)**
33. Dark, C., Ali, N., Golenkina, S., Dhyani, V., Blazeov, R., Parker, B.L., Murphy, K.T., Lynch, G.S., Senapati, T., Millard, S.S., Judge, S.M., Judge, A.R., **Giri, L.**, Russell, S.M., and Cheng, L.Y. 2024. Mitochondrial fusion and altered beta-oxidation drive muscle wasting in a Drosophila cachexia model. *EMBO reports*, pp.1-24. **Impact factor: 8.8 (Q1)**
34. Das, S., **Giri, L.**, & Majumdar, S. (2024). Interaction-based perspective for designing polymer biomaterial: a strategic approach to the chitosan-glycerophosphate system. *ACS biomaterials science & engineering*, 10(7), 4359-4373. **Impact factor: 5.5 (Q1)**

## 2025

35. Saha, D., Pandillapally, R., Dhyani, V., Harsha, K. S., Swain, S., Duraiswamy, S., & **Giri, L.** (2025). Two-step fabrication of clear view SLA millifluidic device for long-term in-vitro cultures. *Rapid Prototyping Journal*, 31(2), 316-326. **Impact factor 3.6 (Q2)**
36. Panigrahi, S. K., Saha, D., Das, S., Majumdar, S., & **Giri, L.** (2025). Formulation and In Vitro Assessment of a Hybrid Gel-Microparticle Composite for Ocular Delivery of Dual-Antiglaucoma Drugs. *Macromolecular Bioscience*, e00270. **Impact factor 4.1 (Q1)**
37. Saha, D., Kumar, S., Gupta, R., Kaur, I., Majumdar, S., Karunarathne, A., **Giri, L.** (2025). Role of calcium in the regulation of chronic stress-induced progression of stress granule assembly in N2a cells. *ACS Chemical Neuroscience*, **Impact factor 3.9 (Q1)**
38. Mahadevan, J., Mitra, K., & **Giri, L.** (2025). An Overview of Virus-Free Protein Expression in Insect Cells: A Mode of Rapid Manufacturing Platform for Therapeutic Protein and Virus-Like-Particles. *Biotechnology and Bioengineering*. **Impact factor 3.6 (Q2)**

## Book chapter

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39. V. Dhyani, A. S. Chann, **L. Giri**, S. M. Russell, and M. Charnley, "A pipeline for dynamic analysis of mitochondrial content in developing T cells: bridging the gap between high throughput flow cytometry and single-cell microscopy analysis", In *Methods in Molecular Biology*, Springer Nature, 2024.
40. V. Dhyani, S. Swain, R.K. Gupta, A. Saxena, R. Singh, **L. Giri**, "Role of Metabotropic Glutamate Receptors (mGluRs) in the Regulation of Cellular Calcium Signaling: Theory, Protocols, and Data Analysis." In: *Metabotropic Glutamate Receptor Technologies*, Neuromethods, vol. 164, Springer Nature, 2021.
41. D. Saha, D., R. K Gupta., I. Kaur, S. Majumdar, & **L. Giri**, "Single cell resolution imaging techniques in optic nerve models in vivo and in vitro." In *Proteomics, Multi-Omics and Systems Biology in Optic Nerve Regeneration* (pp. 57-67). Academic Press, 2025.
42. J. Mahadevan, K. Mitra, and **L. Giri**. "Nonviral Platform for Expression of Recombinant Protein in Insect Cells." In *Baculovirus: Methods and Protocols*, pp. 289-300. Springer US, 2024.

## Patents

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### Sanctioned:

1. Sarpras Swain, Lopamudra Giri, Suryakumar, Falguni Pati, Mrityunjay Suar. **Microfluidic devices and methods of fabrication thereof**, Indian Patent (Application number: 201941027664, Filing date: 10<sup>th</sup> July 2019, Accepted, 9<sup>th</sup> April 2021).
2. Lopamudra Giri, Sayantan Ghosh, Aditya Gupta, **Kit for detecting a protein specific to diabetic retinopathy**, Indian Patent, (Application number: 202441037060 Filing date: 10<sup>th</sup> May 2024, Accepted 25<sup>th</sup> June, 2025).

### Filed:

3. Mahadevan, J., Giri, L. and Mitra, K., 2024, **A biocompatible head plate for semi batch bioreactors**, Indian patent applied (Application number: 202441035701, Filing Date: 10<sup>th</sup> May 2024)

## Technology developed, transferred, startups, if any from IITH

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### *Fabrication of microfluidic/millifluidic/glass-bottom imaging device*

1. **Drug toxicity testing through imaging:** Indirect fabrication of microfluidic device (Fused deposition modelling) using 3D printed template suitable for mammalian cell culture and imaging using confocal microscopy. (Indian Patent granted: Application number: 201941027664, “Microfluidic devices and methods of fabrication thereof”, TRL 4).
2. **Co-culture based drug screening:** Stereolithography-based two-step fabrication for rapid prototyping of clear view millifluidic device for long-term in-vitro co-culture of neuron and glial cells. The device can be used for screening potential drugs for treatment of neurodegeneration. (Prototype tested, TRL-3).  
Currently we are in conversation with Centre for Predictive Human Model Systems (CPHMS), Atal Incubation Centre, CCMB, Hyderabad, for testing these technologies (1 and 2) for generation of non-animal models (NAMs) for neurodegeneration.
3. **Disease specific protein detection (VEGF and IL6):** Fabrication of polymer coated glass- bottom PDMS device for imaging disease specific protein in human serum (portable and affordable). This is designed towards low-cost portable diagnostic kit at rural setting.  
(Indian Patent granted: Application number: 202441037060A: “Kit for detecting a protein specific to diabetic retinopathy”, TRL 4). Currently, we have applied for the **Bharat Innovates,2026, Healthcare**, IIT Bombay to get mentorship for transferring the technology to the market.

## Teaching Experience

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### **Chemical Engineering: Undergraduate core courses**

1. Reaction Engineering (2014-2016)
2. Design of Transport process (2014-2016)
3. Liquid-liquid separation (2017-2019)
4. Biomolecular kinetics (2017-2019)
5. Cell dynamics (2017-2019)
6. Separation II (2020-2022)
7. Mass transfer II (2023-2025)
8. Heterogeneous Catalysis (2017-2020)

### **New courses developed in Chemical Engineering Department [Elective Courses]**

1. Biochemical signals and Systems (2014-2016)
2. Statistical Methods for experimental research (2017-2019)
3. Data Analysis Tools for experimental Research (2020-2024)
4. Statistical computing (2019-2020)
5. Computer vision for chemical and life sciences (2025, Very Recent)

### **Polymer and Biosystems Engineering, M Tech Core courses**

1. Systems Biology I (2020-2024)
2. Systems Biology II (2020-2024)
3. Physical Biology II (2020-2023)

## **Administrative Activities at IITH (from 2013 to till date)**

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### **Institute Level**

- Heading the collaboration plans between Amgen, Hyderabad -IIT Hyderabad (2025)
- Chair, Women Cell, (2017-2020)
- Constitution committee, of Institute biosafety Committee (2015)
- Biosafety committee member (2016-2021)
- Co-ordinator for the M tech program on “Polymers and Biosystems Engineering”, Center for Interdisciplinary Program at IITH, (2022-2023)
- Recruitment committee for Technical Superintendent/Junior Technician, for Chemical/Biomedical Department as well as animal facility at IIT Hyderabad (2017-2020)
- Internal promotion committee for Technical Superintendent and Junior Technician, All departments, (2016-2019).
- UG Student mentoring committee (Students having backlogs and low grades) (2017-2019).

### **Department level**

- Chair, Student Welfare and Alumni Engagement, ChE@IITH (Oct 2025 to till Date).
- Placement Co-ordinator, ChE@IITH, (Nov 2020 to Oct, 2025):  
**Uniqueness:** Implemented a basket-wise placement strategy to improve on the percentage placement as well as average salary per annum (LPA). In this direction, students got placed in several baskets including core Chemical, software, AI/ML and consultancy job along with higher studies. Also, we arranged training programs/mentoring sessions for the B Tech students by the Alumni of Chemical Engineering.
- Alumni Engagement Coordinator, ChE@IITH , (Nov 2020 to Oct, 2025):  
We have started an alumni talk series from various areas like Energy, Sustainability, HealthCare from both Industry/Academia.
- Department Faculty Search Committee member, ChE@IITH (November 2022 to Oct 2025).
- Department Space Committee member, ChE@IITH, (Oct 2020 to till Date).
- Department Staff Grievance committee member, ChE@IITH, (2020-2023).
- Officiating HOD, ChE@IITH, (on several occasions in 2024-2025).
- Faculty advisor, B Tech Batch, ChE@IITH, (2014-2018).
- Timetable committee member, ChE@IITH , (2013-2016).
- Organization of Department Day and other events, ChE@IITH, (2015-2018).

### **Workshop/Conclave (Invited participant)**

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1. ***Emerging Science Technology and Emerging conclave (ESTIC)***, New Delhi, Bharat Mandapam, 2-5,2025.
2. ***Unifying theories in High dimensional Biophysics***, ICTS, Bangalore, July 21 to Aug 1, 2025.  
Organizer: Marianne Bauer (Delft University of Technology, Netherlands), Akshit Goyal (ICTS-TIFR, India), Sidhartha Goyal (University of Toronto, Canada) and Gautam Reddy (Princeton University, USA).

### **International Conferences (presentations)**

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1. *IEEE EMBS Neural Eng. Conf. (NER)*, 2016, Montpelliar, Fance, 2015
2. *Micro and Nanotechnol. Med. Conf.*, Kauai, Hawaii, 2018.
3. *IEEE EMBS Neural Eng. Conf. (NER)*, 2019, San Francisco, USA.
4. *IEEE Eng. Med. Biol. Soc. (EMBC), Engineering in Medicine and Biology*, 2022, Glasgow, UK
5. *IEEE EMBS Neural Eng. Conf. (NER)*, 2023, Baltimore, USA
6. *IEEE Eng. Med. Biol. Soc. (EMBC), Engineering in Medicine and Biology*, 2024, Orlando, USA
7. *ICSB, International conference on Systems Biology*, 2024, Mumbai, India.

## Reviewing

**Biology/Neuroscience:** ACS Neuroscience, Nature Scientific report, STAR Protocols, Applied Microbiology and Biotechnology, Journal of medicinal Chemistry, PLOS ONE, Integrative Biology, iScience.

**AI/ML:** Computers in Medicine and Biology, Biomedical signal processing and control, IEEE Journal of Engineering in Medicine and Biology,

**Computational Biology:** Physica Scripta, Communications in non-linear science and numerical simulation, Heliyon, Neural Engineering.

**Materials:** Colloid and interface science communications.

## National Level responsibilities

**Organizing conference:** CHEMREFERENCE, IIT Hyderabad, India. Organizing Committee, (2015, Dec).

**Grant review:** SERB, Grant review (2016-Till date).

ANRF Grant– Grant Review (2025).

MoE-STARS Biological Sciences, Grant review, (2023).

**Journal review:** The Institution of Engineers (India): Series E, Transactions of the Indian National Academy of Engineering, (2019-Till date).

**Session Chair:** IEEE Applied Sensing Conference, India, IIT Hyderabad, Chairing a session of Medical and Digital Healthcare, 22<sup>nd</sup> Jan 2025.

## Professional membership

1. Member, IEEE, Engineering in Medicine and Biology Society
2. Life Member, Society for Alternatives to Animal Experiments

## Establishment of *in situ* imaging (Live cell imaging) and Data Analysis Lab at IITH [2014-2025]

|   | Description of facility                                                                         | Year                                                                                                          | Year      | Role                                                                                                                                                                                           |
|---|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Live cell imaging facility.<br><br>Facility for in situ Imaging under hypoxia and normoxia.     | Laser scanning Confocal Microscopy (Leica Sp8) attached with incubation chamber and gas controlling facility. | 2014-2017 | Proposal writing, planning and designing of the facility (rooms) suitable for fluorescent imaging, cell culture, bioprinting, procurement of the instrument and establishment of the facility. |
| 2 | Dark room set up.                                                                               | Fluorescent Microscopy (Nikon, Eclipse, Ti2 Main), Digital Sight 50 M Microscope Camera).                     | 2025      |                                                                                                                                                                                                |
| 3 | Mammalian cell culture facility (Non-animal model , NAMS, Cell lines, primary cells, ex vivo) . | Biosafety hood (2), Separate hoods are used for in vitro and ex vivo.                                         | 2015-1018 |                                                                                                                                                                                                |
| 4 | Biochemical assays facility for measurement small molecules, protein, and RNA.                  | ELISA reader, Nanodrop (Estimation of concentration in small volume)                                          | 2023      |                                                                                                                                                                                                |
| 5 | Fermenter Facility with controllers and sensors (oxygen and pH).                                | Bioreactor (3L) for optimization of protein production                                                        | 2015      |                                                                                                                                                                                                |
| 6 | Milifluidic device Fabrication facility.                                                        | Stereolithography based 3D Printer for printing templates<br>Plasma cleaner.                                  | 2019-2025 |                                                                                                                                                                                                |
| 7 | Bioprinting facility.                                                                           | Extrusion 3D printer (Very recent)                                                                            | 2025      |                                                                                                                                                                                                |
| 8 | Computing facility for AI/ML based analysis of videos/images                                    | Computers and clusters                                                                                        | 2020-2025 |                                                                                                                                                                                                |

Live Cell imaging facility: Confocal microscopy attached to incubator (controlled temperature and CO<sub>2</sub> level), Dark room for fluorescent imaging. (First in IITH: in Chemical Engineering department)



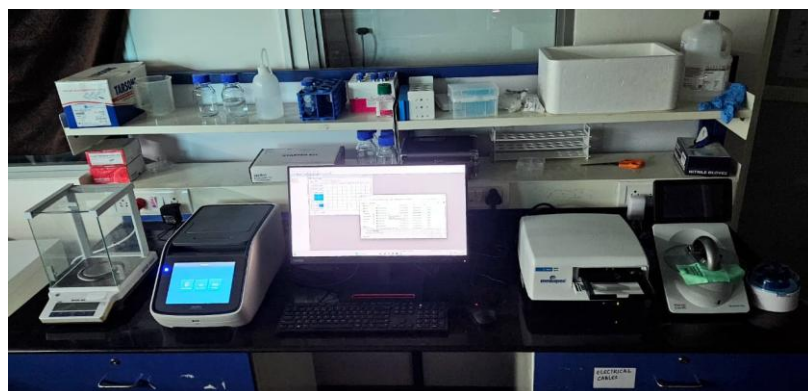
Mammalian Cell culture facility: Biosafety level 2



Bioprinting facility



Analytical biochemistry facility



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**Collaboration with L V Prasad Eye hospital:** Our research output is being used to evaluate various drugs for Glaucoma and Diabetic retinopathy. We have obtained multiple grants from DBT and ICMR with L V Prasad Eye hospital and have multiple joint publications. Currently, we have applied a joint proposal for Center of Excellence (IITH-L V Prasad Eye Hospital) in a call from ICMR.

**Projects (sponsored/Consultancy) [Ongoing and Completed]**

| Sl. No. | As PI or Co-PI | Project Title                                                                                                                                                                              | Sponsoring Agency                                                                                                 | Period of project      | Project Cost                   |
|---------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------|--------------------------------|
| 1       | PI             | A vision for vision: A program to create an opportunity to study biology towards clinical application of novel gene therapy.<br>Host for Vaishvik Bharatiya Vaigyanik (VAIBHAV) Fellowship | Department of Science and technology, Science and Engineering Research Board (SERB), New Delhi.                   | 2025-2028<br>(ongoing) | 72.22 lakh (Rs)                |
| 2       | PI             | Portable imaging solutions towards affordable healthcare: Microfluidic devices for the detection of disease-specific proteins                                                              | Royal Academy Engineering, Engineering X, UK                                                                      | 2023-2026<br>(ongoing) | 65000 SP +43000 SP (Follow on) |
| 3       | PI             | Sustained dual drug release of Timolol Maleate and Latanoprost for Glaucoma Treatment: In vivo studies towards cross-linker free drug-polymer formulation for usage as eye drop            | Indian Council of Medical Research, Government of India.                                                          | 2025-2028<br>(ongoing) | 380.47 lakh (Rs)               |
| 4       | PI             | Investigation on the biophysical factors regulating the phase transition mechanisms for assembly and disassembly of stress granules                                                        | Department of Biotechnology, Government of India, New Delhi.                                                      | 2023-2026<br>(ongoing) | 85.24 lakh (Rs)                |
| 5       | PI             | Development of computational and visualization software for evaluating GPCR targeting drugs with the aim of mitigating coronavirus infection level”, Department of Science and Technology  | Mathematical Research Impact-centric Support (MATRICS), Science and Engineering Research Board (SERB), New Delhi. | 2020-2021              | 5 lakh (Rs)                    |
| 6       | PI             | Multiple-Drug Nano-formulation and Quantitative Imaging of Cellular Response for Glaucoma Treatment Protocol (Collaborative Project with L V Prasad Eye Hospital).                         | Department of Biotechnology, Government of India, New Delhi.                                                      | 2019-2022              | 87 lakh (Rs)                   |
| 7       | PI             | Miniaturized bioimaging device with embedded nano sensor for real-time oxygen sensing                                                                                                      | Department of Biotechnology, Government of India, New Delhi.                                                      | 2018-2021              | 65.10 lakh (Rs)                |
| 8       | PI             | The role of intracellular calcium signaling in matrix differentiation of mesenchymal stem cells                                                                                            | Department of Biotechnology, Government of India, New Delhi.                                                      | 2018-2021              | 39.97 lakh (Rs)                |
| 9.      | PI             | Integrated approach towards economic production of baculovirus pesticide: Combination of genetic engineering, systems biology and bioreactor design”                                       | Science and Engineering Research Board (SERB), Government of India, New Delhi, Young Scientist.                   | 2015-2018              | 25.4 lakh (Rs)                 |
| 10      | Co-PI          | Application of artificial intelligence and optimal control in optimizing protein production in bioreactor system: A step towards affordable biomolecule production                         | Department of Biotechnology, Government of India, New Delhi.                                                      | 2020-2024              | 65 lakh (Rs)                   |
| 11      | Co-PI          | Investigation on role of neuron-glia activation leading to neuronal loss as an                                                                                                             | Science and Engineering Research Board (SERB),                                                                    | 2018-2021              | 46.37 lakh (Rs)                |

|                                                                                                       |       |                                                                                                                                                                                                 |                                                                                                        |           |                  |
|-------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------|------------------|
|                                                                                                       |       | early marker of diabetic retinopathy using transcriptomics and high-resolution imaging-based approaches, Collaborative Project with L V Prasad Eye Hospital).                                   | Government of India, New Delhi.                                                                        |           |                  |
| 12                                                                                                    | Co-PI | An ultra-sensitive metal oxide nano-fiber platform for the early detection of MMP2, MMP9 and C3 proteins from the tear fluid of preterm infants for the diagnosis of Retinopathy of Prematurity | Science and Engineering Research Board (SERB), Impacting research, innovation and Technology (IMPRINT) | 2020-2023 | 118.96 lakh (Rs) |
| 13                                                                                                    | Co-PI | Development of computational software integrating multilevel image data analysis: Towards efficient clinical practices and advanced biomolecular research in ophthalmology".                    | Department of Biotechnology, Government of India, New Delhi.                                           | 2017-2020 | 36.2 lakh (Rs)   |
| <b>Total amount as PI = INR 760 lakhs + 1,08,000 UK Pound; Total amount as Co-PI = INR 266 lakhs.</b> |       |                                                                                                                                                                                                 |                                                                                                        |           |                  |

#### PhD guidance: List of completed and ongoing PhD students at IITH

|     | PhD Student name                                                                | Status         | Current position                   | Affiliation                                                                                      |
|-----|---------------------------------------------------------------------------------|----------------|------------------------------------|--------------------------------------------------------------------------------------------------|
| 1.  | Koushik Maparu                                                                  | Ongoing        | PhD Scholar                        | IITH                                                                                             |
| 2.  | Samudra Bhowmik (Co-Supervisor-Dr. Saptarshi Majumdar)                          | Ongoing        | PhD Scholar                        | IITH-DEAKIN joint PhD Program                                                                    |
| 3.  | Shrirum Bhutt B Co-superviosr – Lopamudra Giri, Supervisor - Saptarshi Majumdar | Ongoing        | PhD Scholar                        | IIT Hyderabad                                                                                    |
| 4.  | Shivani Chintala                                                                | Ongoing        | PhD Scholar                        | IITH                                                                                             |
| 5.  | Sayantan Ghosh                                                                  | Ongoing        | PhD Scholar                        | IITH                                                                                             |
| 6.  | Satya Deepika Neelapala                                                         | Ongoing        | PhD Scholar                        | IITH                                                                                             |
| 7.  | Jagdeesh Mahadevan (Co-Supervisor-Dr. Kishalay Mitra)                           | Ongoing        | PhD Scholar                        | IITH-Swinburne Institute of Technology joint PhD program                                         |
| 8.  | Dr. Debasmita Saha (2020-2025) (Co-Supervisor-Dr. Saptarshi Majumdar)           | Degree Awarded | Post Doctoral Fellow               | Indian Institute of Science Education and Research Tirupati                                      |
| 9.  | Dr. Vaibhav Dhyani (2018-2023)                                                  | Degree Awarded | Postdoctoral Fellow                | The Herbert Wertheim UF Scripps Institute for Biomedical Innovation & Technology, Florida , USA. |
| 10. | Dr. Suman Gare (2017-2022)                                                      | Degree awarded | Postdoctoral Fellow                | Institute of Experimental Oncology, University Hospital Bonn, Germany                            |
| 11. | Dr. Manohar Kuruba (DST Inspire Fellow) (2017-2022)                             | Degree awarded | Assistant Professor, Biotechnology | Bharatiya Engineering Science & Technology Innovation University, Andra Pradesh                  |

|     |                                                                                         |                |                      |                                                                                                  |
|-----|-----------------------------------------------------------------------------------------|----------------|----------------------|--------------------------------------------------------------------------------------------------|
| 12. | Dr Abha Saxena, (DST INSPIRE Fellow) (2015-2021)                                        | Degree Awarded | Lead Scientist       | Metflux Research, Bangalore.                                                                     |
| 13. | Dr. Sarpras Swain (2014-2019)                                                           | Degree Awarded | Research Scientist   | Reno School of Medicine, University of Nevada, Reno.                                             |
| 14. | Dr. Saugat Das<br>Co-superviosr –<br>Lopamudra Giri, Supervisor -<br>Saptarshi Majumdar | Degree Awarded | Post doctoral Fellow | University of Naples, Federico II, Chemical and Materials Engineering and Industrial production. |

## Publications (International/ National Conferences, Scopus Indexed)

### 2015

1. P. Reddy, S. Shukla, A. Karunarathne, S. Jana, and **L. Giri**, "Segmentation of neurons and measurement of optically programed neurite growth: Fast automation via Bayesian thresholding," in *Proc. 7th IEEE EMBS Neural Eng. Conf. (NER)*, 2015, pp. 996–999, doi: 10.1109/NER.2015.7146794.

### 2016

2. J. G. Joseph, S. Nanda, S. Vennamalla, G. Bhure, R. Singh, S. Jana, and **L. Giri**, "Integrated quantification based on confocal imaging: Cell crowding modulates heterogeneity in GPCR-mediated calcium oscillation," in *Proc. 38th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2016, pp. 3891–3894, doi: 10.1109/EMBC.2016.7591578.

### 2017

3. S. Swain, S. Ande, R. Suryateja, S. Jana and **L. Giri**, "Spatially resolved calcium spiking in hippocampal neurons: Estimation via confocal imaging and model-based simulation," in *Proc. 8th IEEE EMBS Neural Eng. Conf. (NER)*, 2017, pp. 279–283, doi: 10.1109/NER.2017.8008345.

### 2018

4. S. Sahu, G. V. P. Kumar, B. Paul, R. Singh, S. Jana, **L. Giri**, and S. G. Singh, "Towards early detection of hypoxia: Fabrication of facile, robust, cobalt doped ZnO nanofiber for in situ sensing of dissolved oxygen," in *Proc. Micro and Nanotechnol. Med. Conf.*, 2018.
5. S. Soumitri, P. D. Pantula, **L. Giri**, and K. Mitra, "Smart data analytics approach to model complex biochemical oscillations in hippocampal neurons," in *Proc. 40th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2018, pp. 5045–5048, doi: 10.1109/EMBC.2018.8513414.
6. S. Swain, P. D. Pantula, K. Mitra, and **L. Giri**, "Confocal imaging of cytosolic  $\text{Ca}^{2+}$  and fuzzy clustering reveal the circuit topology details underlying synchronization in hippocampal neurons," in *Proc. 40th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2018, pp. 1–4, doi: [10.1109/EMBC.2018.8512433](https://doi.org/10.1109/EMBC.2018.8512433)

### 2019

7. R. Singh, A. Saxena, and **L. Giri**, "Single neuron imaging reveals metabotropic glutamate receptor mediated bursting and delay in calcium oscillation in hippocampal neurons," in *Proc. 41st Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2019, pp. 5146–5149, doi: [10.1109/EMBC.2019.8856638](https://doi.org/10.1109/EMBC.2019.8856638).
8. A. Saxena, V. Dhyani, V. Gare, and **L. Giri**, "Effect of topology and time window on probability distribution underlying baclofen induced  $\text{Ca}^{2+}$  response in hippocampal neurons," in *Proc. 41st Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2019, pp. 2997–3000, doi: [10.1109/EMBC.2019.8857601](https://doi.org/10.1109/EMBC.2019.8857601).
9. V. Upadhyay, S. T. Ravutla, V. Dhyani, K. George, S. Swain, K. Mitra, and **L. Giri**, "A model screening framework for the generation of  $\text{Ca}^{2+}$  oscillations in hippocampal neurons using differential evolution," in *Proc. 9th IEEE EMBS Neural Eng. Conf. (NER)*, 2019, pp. 961–964, doi: [10.1109/NER.2019.8716993](https://doi.org/10.1109/NER.2019.8716993).
10. D. Spoorthy, S. R. Manne, V. Dhyani, S. Swain, S. Shahulhameed, S. Mishra, I. Kaur, **L. Giri**, and S. Jana, "Automatic identification of mixed retinal cells in time-lapse fluorescent microscopy images using high-

dimensional DBSCAN,” in *Proc. 41st Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2019, pp. 4783–4786, doi: 10.1109/EMBC.2019.8857375.

11. A. Saxena, V. Upadhyay, V. Dhyani, S. Jana and **L. Giri**, "Cell-to-Cell Variability in Protein Expression during Viral Infection: Monte-Carlo Simulation and Validation based on Confocal Imaging,” in *Proc. 41st Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, , 2019, pp. 138-141, doi: 10.1109/EMBC.2019.8856612.
12. P. D. Pantula, S. S. Miriyala, S. Swain, **L. Giri** and K. Mitra, "Automation of Synchronicity Identification in Hippocampal Neurons through Intelligent Data Clustering Approach," *6th Indian Control Conference (ICC)*, Hyderabad, India, 2019, pp. 268-273, doi: 10.1109/ICC47138.2019.9123194.
13. R. Singh, S. Sharma, **L. Giri** and K. Mitra, "Dynamic optimization for Optimal production of Indole in a mixed culture," *6th Indian Control Conference (ICC)*, Hyderabad, India, 2019, pp. 256-261, doi: 10.1109/ICC47138.2019.9123170.

## 2020

14. S. Chel, S. Gare, and **L. Giri**, “Detection of specific templates in calcium spiking in HeLa cells using hierarchical DBSCAN: Clustering and visualization of cell-drug interaction at multiple doses,” in *Proc. 42nd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2020, pp. 2425–2428, doi: [10.1109/EMBC44109.2020.9175925](https://doi.org/10.1109/EMBC44109.2020.9175925).
15. A. Saxena, V. Dhyani, S. Jana, and **L. Giri**, “Application of Kohonen self-organizing map to cluster drug-induced  $\text{Ca}^{2+}$  response in hippocampal neurons at different drug dose,” in *Proc. IEEE Int. Symp. Biomed. Imag. (ISBI)*, 2020, pp. 1–4, doi: [10.1109/NCC48643.2020.9056031](https://doi.org/10.1109/NCC48643.2020.9056031).
16. P. D. Pantula, S. S. Miriyala, **L. Giri**, and K. Mitra, “Synchronicity identification in hippocampal neurons using artificial neural network assisted fuzzy c-means clustering,” in *Proc. IEEE Symp. Comput. Intell. (SSCI)*, 2020, pp. 1594–1600, doi:[10.1109/SSCI47803.2020.9308344](https://doi.org/10.1109/SSCI47803.2020.9308344).
17. G. Suman, A. Saxena, and **L. Giri**, “Confocal imaging of intercellular calcium in HeLa cells for monitoring drug response: Biophysical framework for visualization of time-lapse images,” in *Proc. IEEE Int. Symp. Biomed. Imag. (ISBI)*, 2020, pp. 1–4, doi: [10.1109/ISBI45749.2020.9098697](https://doi.org/10.1109/ISBI45749.2020.9098697).

## 2021

18. S. Sharma, K. N. Pujari, S. S. Miriyala, **L. Giri**, and K. Mitra, “Unstructured modeling and RNN surrogate development for optimizing vaccine production in baculovirus expression vector system,” in *Proc. 7th Indian Control Conf. (ICC)*, 2021, pp. 406–411, doi: 10.1109/ICC54714.2021.9703114.
19. S. Sharma, A. Saxena, S. Chel, K. Mitra, and **L. Giri**, “Mathematical modeling of viral infection dynamics and immune response in SARS-CoV-2: A computational framework for testing drug efficacy,” in *Proc. 43rd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2021, pp. 4370–4373, doi: 10.1109/EMBC46164.2021.9630629.
20. S. Gare, S. Chel, M. Kuruba, S. Jana, and **L. Giri**, “Dimension reduction and clustering of single cell calcium spiking: Comparison of t-SNE and UMAP,” in *Proc. Nat. Conf. Commun. (NCC)*, 2021, pp. 1–6, doi: 10.1109/NCC52529.2021.9530128.
21. S. Ande, S. Avasarala, A. Karunarathne, **L. Giri**, and S. Jana, “Statistical dependence between neuronal spike train pairs: Quantification based on empirical mutual information rate,” in *Proc. IEEE EMBS Int. Conf. Biomed. Health Informat. (BHI)*, 2021, pp. 1–4, doi: 10.1109/BHI50953.2021.9508584.
22. V. Dhyani, S. Jana, and **L. Giri**, “Gaussian mixture modeling of single-neuron responses obtained from confocal calcium imaging of dissociated rat hippocampal neurons,” in *Proc. 10th IEEE EMBS Neural Eng. Conf. (NER)*, 2021, pp. 145–148, doi: 10.1109/NER49283.2021.9441102.
23. S. Ande, J. R. Regatti, N. Pandey, A. Karunarathne, **L. Giri**, and S. Jana, “Heterogeneity in neuronal calcium spike trains based on empirical distance,” in *Proc. 10th IEEE EMBS Neural Eng. Conf. (NER)*, 2021, pp. 141–144, doi: 10.1109/NER49283.2021.9441175.

24. S. Ande, S. Avasarala, J. R. Regatti, N. Pandey, S. Swain, A. Karunarathne, **L. Giri**, and S. Jana, "Information content in neuronal calcium spike trains: Entropy rate estimation based on empirical probabilities," in *Proc. IEEE EMBS Conf. Biomed. Eng. Sci. (IECBES)*, 2021, pp. 103–108, doi: 10.1109/IECBES48179.2021.9398798.

## 2022

25. S. Sharma, **L. Giri**, and K. Mitra, "Multi-objective optimization and control under uncertainty for performance improvement of a baculovirus expression vector system," in *Proc. 8th Indian Control Conf. (ICC)*, 2022, pp. 416–421, doi: 10.1109/ICC56513.2022.10093623.
26. S. Gare, S. Chel, P. D. Pantula, A. Saxena, K. Mitra, R. Sarkar, and **L. Giri**, "Analytics pipeline for visualization of single cell RNA sequencing data from bronchoalveolar fluid in COVID-19 patients: Assessment of neuro fuzzy-c-means and HDBSCAN," in *Proc. 44th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2022, pp. 1634–1637, doi: 10.1109/EMBC48229.2022.9871686.

## 2023

27. D. Saha, V. Dhyani, and **L. Giri**, "In vitro laser scanning confocal microscopy and unsupervised segmentation: Quantification of cytosolic calcium and RNA distribution in hypoxic neurons," in *Proc. 45th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2023, pp. 1–4, doi: 10.1109/EMBC40787.2023.10340952.
28. D. Saha, S. Hadule, and **L. Giri**, "A deep learning approach for automation in neurite tracing and cell size estimation from differential contrast images under healthy and hypoxic conditions," in *Proc. 45th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2023, pp. 1–4, doi: 10.1109/EMBC40787.2023.10340948.
29. D. Sehanobish, V. Dhyani, S. Gare, K. Mitra, and **L. Giri**, "Microstate analysis of GABA B and mGluR mediated modulation of calcium spiking in hippocampal neurons," in *Proc. 11th IEEE EMBS Neural Eng. Conf. (NER)*, 2023, pp. 1–4, doi: 10.1109/NER52421.2023.10123717.
30. V. Dhyani, K. V. Venkatesh, and **L. Giri**, "Computational framework for in silico analysis of neural hyperactivity and loss of neural activity in a population of interconnected neurons," in *Proc. 11th IEEE EMBS Neural Eng. Conf. (NER)*, 2023, pp. 1–4, doi: 10.1109/NER52421.2023.10123884.
31. S. D. Neelapala, S. Gare, S. Jana, and **L. Giri**, "Multi-frame sampling and DBSCAN based approach for segmentation of HeLa cells from time-lapse fluorescent images," in *Proc. IEEE 20th India Council Int. Conf. (INDICON)*, 2023, pp. 776–781, doi: 10.1109/INDICON59947.2023.10440772.
32. V. S. Masampally, S. Sharma, **L. Giri**, and K. Mitra, "Physics informed neural networks for baculovirus–insect cell system," in *Proc. 9th Indian Control Conf. (ICC)*, 2023, pp. 22–27, doi: 10.1109/ICC61519.2023.10442232.

## 2024

33. S. D. Neelapala, S. Gare, V. Dhyani, D. Srikanth, S. Jana, and **L. Giri**, "Improved segmentation of confocal calcium videos of HeLa cells using deep-learning assisted watershed algorithm," in *Proc. 46th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2024, pp. 1–4, doi: 10.1109/EMBC53108.2024.10781812.
34. D. Srikanth, J. Gurung, S. D. Neelapala, V. Joshi, **L. Giri**, P. Vaddavalli, and S. Jana, "Next-generation teleophthalmology: AI-enabled quality assessment aiding remote smartphone-based consultation," in *Proc. 46th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2024, pp. 1–4, doi: 10.1109/EMBC53108.2024.10782045.
35. J. Mahadevan, N. Bogoria, S. D. Neelapala, D. Srikanth, S. Jana, K. Mitra, and **L. Giri**, "Towards fluorescent-tag-less viral titration: Automated estimation of cell-size distribution and infection level from phase-contrast microscopy using deep learning and transfer learning," in *Proc. 46th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2024, pp. 1–4, doi: 10.1109/EMBC53108.2024.10782022.
36. A. Mallick, A. Shaiju, S. D. Neelapala, **L. Giri**, R. Sarkar, and S. Jana, "AI-based 3-lead to 12-lead ECG reconstruction: Towards smartphone-based public healthcare," in *Proc. IEEE Int. Conf. e-Health Netw., Appl. Serv. (HealthCom)*, 2024, pp. 1–6, doi: 10.1109/HealthCom60970.2024.10880752.

37. S. D. Neelapala, S. Jana, and **L. Giri**, “U-Net-based HeLa cell segmentation with zero manual labeling using DBSCAN-generated annotations,” in *Proc. IEEE Int. Conf. e-Health Netw., Appl. Serv. (HealthCom)*, 2024, pp. 1–3, doi: 10.1109/HealthCom60970.2024.10880723.
38. V. S. Masampally, S. Sharma, **L. Giri**, and K. Mitra, “Physics informed deep operator networks for baculovirus–insect cell system,” in *Proc. 10th Indian Control Conf. (ICC)*, 2024, pp. 514–519, doi: 10.1109/ICC64753.2024.10883680.

## 2025

39. A. Amal, S. Ghosh, A. Gupta, S. D. Neelapala, S. Ghosh, R. Sarkar, K. Mitra, H. Narasimhan, S. Jana, and **L. Giri**, “Novel lightweight SDR-UNet model enables fast and accurate detection of interleukin-6 based on portable fluorescent imaging,” in *Proc. IEEE Int. Conf. e-Health Netw., Appl. Serv. (HealthCom)*, 2025, pp. 1–4.
40. S. D. Neelapala, S. Gare, V. Dhyani, K. Mitra, S. Jana, and **L. Giri**, “Biochemical oscillations in HeLa cells: Automated time-lapse image analysis and two-step parameter estimation,” in *Proc. 47th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2025, pp. 1–4.
41. A. K. Singh, S. D. Neelapala, M. Venkatesh, V. Dhyani, S. Jana, S. Majumdar, and **L. Giri**, “Quantification of stress granules under dynamic calcium-induced stress using deep learning-based two-step segmentation”, in *Proc. 11th IEEE EMBS Neural Eng. Conf. (NER)*, 2025, pp. 1–4.

## Conference Presentations

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1. J. Mahadevan, K. Mitra, L. Giri, H. Sumer, S. Moulton, “Surface modification of PDMS by plasma polymerization to support insect cell culturing for recombinant protein expression” in **29th annual ASBTE conference 2025, Brisbane**, Australia, April 20-25, 2025.
2. Debasmita Saha, Lopamudra Giri, "An imaging method to study the stress granule formation mediated neurite disintegration during late phase of chronic hypoxia", **Longevity India Conference, IISC Bangalore**, 2025.
3. Saloni Singh, and Megha Venkatesh, Live cell imaging of stress granules for monitoring of neuronal aging under intermittent stress, Longevity India Conference, IISC Bangalore, 2025.
4. S.D. Neelapala, S.Gare and L. Giri, “Multi-cell modeling of collective calcium oscillation using machine generated network topology from confocal dataset” 23<sup>rd</sup> **International Conference on Systems Biology, Mumbai**, Dec, 2024.
5. S.D. Neelapala, S.Gare and L. Giri, “Exploring norepinephrine induced calcium Dynamics at the Single-Cell Level: Parameter estimation using GA for various cell states in HeLa cell population”, 23<sup>rd</sup> **International Conference on Systems Biology, Mumbai**, Dec, 2024.
6. S. Bhaumik, S.S.Narayanan, D.Saha, S.Majumdar and L.Giri, “Investigation on biophysical factors controlling the stress granule distribution in a cell population”, **23<sup>rd</sup> International Conference on Systems Biology, Mumbai**, Dec, 2024.
7. B. Pavan Kishore, S.D.Neelapala and L.Giri, “Classification of drug induced calcium oscillation using Long Short-Term Memory (LSTM) network”, 23<sup>rd</sup> International Conference on Systems Biology, Mumbai, Dec, 2024.
8. S.S.Narayanan, S. Bhaumik, V.Dhyani, K.Mitra, S. Duraiwamy and L.Giri, “Identification of calcium channel dynamics in Hypoxia-Induced microglia cells using dynamic optimization”, 23<sup>rd</sup> **International Conference on Systems Biology, Mumbai**, Dec, 2024.
9. A.K.Singh, M.Venkatesh, S. Bhaumik, S.D.Neelapala, S.Majumdar and L.Giri, “Computer vision-based readouts for analysis of stress granules in neuronal cell population: Comparison of protein aggregate size distribution under acute and chronic stress”, 23<sup>rd</sup> **International Conference on Systems Biology, Mumbai**, Dec, 2024.
10. Debasmita Saha, Samyak Bahuguna, Lopamudra Giri, "Spatiotemporal detection of mitochondrial ROS in hypoxic neuronal cells via 3D imaging using laser scanning confocal microscopy (LSCM) and YOLO image detection", **Chemference**, BITS Pilani Goa 2023.
11. J. Mahadevan, Mitra, K. and Giri, L., Mitochondrial ROS quantification method for recombinant protein expression in insect cells, **Chemference**, Goa, Sep 2023”.
12. J. Mahadevan, Mitra, K. and Giri, L., Fabrication of silicone-based microparticles for application in high-density cell culture”, **IICHe-Chemcon**, Kolkata, Dec 2023”.

## International collaborations

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### USA

1. Dr. Bikash Patnaik, Paediatric Ophthalmology, University of Wisconsin, Madison.  
Received **Vaishvik Bharatiya Vaigyanik (VAIBHAV) Fellowship** to host Prof Patnaik in India and work on a project, “A vision for vision: A program to create an opportunity to study biology towards clinical application of novel gene therapy.” [ In collaboration with L V Prasad Eye Institute and NISER Bhubaneswar].
2. Dr. Bryan Copits, Washinton University, Saint Louis. Project on “Computational framework for identification of synaptic connection in hippocampal culture” Manuscript under preparation.
3. Dr. Ajith Karunarathne, Saint Louis University.  
Recent Publication: Saha, D., Kumar, S., Gupta, R., Kaur, I., Majumdar, S., Karunarathne, A., Giri, L. (2025). Role of calcium in the regulation of chronic stress-induced progression of stress granule assembly in N2a cells. *ACS Chemical Neuroscience*.

### UK

4. Dr. Rahuldeb Sarkar, King’s college London, UK, Medway NHS Foundation Trust, Dr. Sourabh Ghosh, Loughborough University.  
3D printing-based fabrication of an imaging device for detection of interleukin-6 at ICU in urban/rural/remote setting. (Lopamudra Giri, Kishalay Mitra, Haripriya Narasimhan)  
Received the Award from Royal Academy Engineering, UK, London (65000 (year 1) + 43000 SP (year 2)).

### Australia

5. Dr. Rosanne Guijt, Deakin University, Australia.  
Project: Innovation in Bioengineering: Biocompatible transparent 3D-Printed Device for spheroid culture and mimicking hypoxic condition, towards developing non-animal model.  
IITH-DEAKIN PhD Student, Samudra Bhowmik.
6. Dr. Sarah Russell, Swinburne University of technology,  
Project: Analysis of T cell culture at various stages and mitochondrial imaging using deep learning-based methods, IITH-SUT PhD Student, Vaibhav Dhyani [ Received Best Thesis award in 2024].  
Book Chapter: V. Dhyani, A. S. Chann, L. Giri, S. M. Russell, and M. Charnley, “A pipeline for dynamic analysis of mitochondrial content in developing T cells: bridging the gap between high throughput flow cytometry and single-cell microscopy analysis”, In *Methods in Molecular Biology*, Springer Nature, 2024
7. Dr. Simon Moulton and Dr. Huseyin Sumer, Swinburne University of technology.  
IITH-SUT PhD Student, Jagadeesh Mahadevan.  
Project: Fabrication of polymer microparticles as microcarrier for insect cell culture: Biomedical Engineering, Swinburne University of Technology,
  - a. Engineering PDMS Surfaces via Plasma Polymerisation to Enhance Cell Compatibility and Recombinant Protein Expression in eukaryotic cells (manuscript in preparation).
  - b. Aging Behaviour of Plasma-Polymerized Films: Surface Stability and Functional Retention Over Time (manuscript in preparation).

## National collaborations

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1. Dr. Inderjeet Kaur, Dr. Subhabrata Chakraborty  
LV Prasad Eye hospital  
Development of Dual-Drug formulations for Glaucoma, Construction of non-animal model for diabetic retinopathy, Establishment of hypoxia in primary mixed retinal culture. [Multiple grant and publications].
2. Professor Deepak Kumar Saini, IISc Bangalore (2025)  
Development of non-animal model for lung fibrosis, A platform for drug screening (Recently started).
3. Professor Sudarshan Neogi, IIT Kharagpur, IIT Kharagpur (2016-2023)  
3D, Imaging nanoparticle internalization in cancer cells (publication in AICHE journal).  
Dey, A., Gare, S., Swain, S., Bhattacharya, P., Dhyani, V., **Giri, L.** and Neogi, S., 2022. 3D imaging and quantification of PLL coated fluorescent ZnO NP distribution and ROS accumulation using laser scanning confocal microscopy. *AICHE Journal*, 68(9), p.e17801.

## Collaborations within IITH

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1. **Saptarshi Majumdar**, Chemical Engineering: Imaging condensates in neurons/axons, Monitoring the dynamics of stress granules under stress conditions. This is a project aiming at understanding fundamental basis of condensate formation and its dynamics in stressed condition.
2. **Kishalay Mitra**, Chemical Engineering: Optimization of process parameters for production of protein, Machine learning methods for classification times series data, parameter estimation using Genetic algorithm.
3. **Suhanya Duraiswamy**, Chemical Engineering: Fabrication of microfluidic device for co-culture of neuron and glia, non-animal model for drug screening. This project aims at a translational aspect of non-animal models for drug testing.
4. **Satyavrata Samavedi**, Chemical Engineering Calcium imaging for functional cells grown on Electrospun fibers with various alignments.
5. **Haripriya Narasimhan**, Liberal Arts, Field work on social impact of diagnostic devices and healthcare technologies in rural areas. This project aims at finding the social relevance of the current medical technologies.

## Invited talks

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### After joining IIT Hyderabad

- “Quantitative live cell imaging and Systems biology: Insights into cell states and population response”, Chemical Engineering Seminar Series, **Indian Institute of Science, Bangalore**, April, 2024
- Lecturer on “Imaging in Nanobiotechnology” (India Side) for the Global Initiative of Academic networks (GIAN) (151001B01) program. Nanobiotechnology: Discipline at Crossroads, **Indian Institute of Technology, Kharagpur**. Foreign Faculty: Vuk Uskokovic, University of Illinois, Chicago, USA, Dec, 2015.
- “Life Science Activities at Indian Institute of Technology, Hyderabad”, **Ritsumeikan University in Japan**, to promote active collaboration between IITH and Japan, March 2015.

### Invited Speaker, conference

- “Live imaging of chronic Hypoxia-Induced neurodegeneration in N2A cells: Assessment using confocal microscopy”, Multi-Spectral Radiation Analyses for Materials Innovation Symposium, - From Cultural Heritages to Medical Applications organized by *Shimane University, Matsue, Japan*. Nov, 2025.
- “Investigation of Chronic Hypoxia-Induced neurodegeneration in N2A cells: Assessment of therapeutic strategy”, India Society for alternatives to animal experiments (SAAE-I), organized by CSIR–Centre for *Cellular and Molecular Biology (CCMB)*, *Hyderabad*. October 2025.
- “Confocal imaging in Healthcare”, Deakin-IIT CoE April 15/16 Health Care Materials & Device e-Symposium, Apr 15-16, 2021.

### Invited Speaker, Workshop

- “Quantitative microscopy and ML assisted system modeling of calcium oscillation in cancer cells” Workshop on Computational Oncology: Modeling and Experiments’, organized by Param Hansa Centre for Computational Oncology at IISc Bangalore, IIT Hyderabad, December, 2024.
- Optical Microscopy Workshop - Nikon CoE and SATHI Centre for In-situ and Correlative Microscopy (CISCoM), IIT Hyderabad, March 2025.
- “Live imaging using Confocal Microscopy”, IIT Guwahati, 3<sup>rd</sup> National Workshop on NEMS/MEMS and Theranostic Devices, Centre for Nanotechnology, Indian Institute of Technology, Guwahati, Feb 2017.
- “Confocal Microscopy, Application in live cell imaging”, Workshop on Confocal Laser Scanning Microscopy (CLSM), Fluorescence Microscopy (FM), Image Processing (IP) Techniques and their applications in Life Sciences, Materials Science, and Engineering" GITAM University, Visakhapatnam campus.

### Before joining IIT Hyderabad

- “Towards Large-Scale Continuous Baculovirus Production in Insect Cell Culture: Overcoming Mutations in *fp25k* Baculovirus Gene” *Baculovirus Technology*, Boston, MA, Sep. 2008.
- “Functional analysis of CYP3A5 and CYP2B6 variants”, *Eli Lilly and Company, Drug disposition group*, Indianapolis, IN, Aug. 2007.
- “Overcoming Mutations in AcMNPV *fp25k* Gene: Removal of The Transposon sites from the Gene”, *Center for Biocatalysis and Bioprocessing Annual Conference*, Iowa City, IA, Oct. 2007.

## Workshop/conference organized

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### Workshop/conference/seminar Series organized

1. **Statistics and Machine learning in Life Science (Feb 4-Feb 6, 2025)**, Chemical Engineering Department, IIT Hyderabad, 2025.  
The main objective of this workshop is to educate the students on Stochastic differential equations, Information theory, Optimization and Convolution neural network/autoencoder, generative network like VAE and autoencoder who are working in the area of life science and Bioengineering.
2. **Systems Biology Seminar series (2025 Jan-2021 Feb)**, Chemical Engineering Department, IIT Hyderabad, 2025.  
The main objective is to motivate the students in the field of Systems Biology, data-driven models through discussion sessions.  
Speaker details: Sana Khanum, University of Chicago, USA, Narendra Dixit, IISc Bangalore, K V Venkatesh, IIT Bombay, Shaon Chakraborty, NCBS Bangalore.
3. **Bridging the Gap: Device Fabrication to Portable Diagnostic Solutions (October 21, 2024)**, Indo-UK Workshop, Chemical Engineering Department, IIT Hyderabad, 2025.  
The main objective is to provide a comprehensive perspective to the students on various topics including medical anthropology, Clinical perspectives on Sepsis, Protein sensors, Device fabrication, 3D Printing, and Analysis of imaging using computer vision models.  
Speaker details Medway Hospital, UK, Rahul Deb Sarkar, Loughborough University, Sourabh Ghosh, Haripriya Narasimhan, IIT Hyderabad, Suhanya Duraiswamy, IIT Hyderabad, Kishalay Mitra, IIT Hyderabad. Co-Ordinator- Lopamudra Giri.
4. **Biomedical Imaging and informatics, (March 7- March 10, 2017)**, Technical Education Quality Improvement Program, Government of India (TEQIP) workshop, at IIT Hyderabad, India.  
The main objective of this workshop was to give exposure to engineering concepts from various disciplines to handle challenges in health and biomedical sciences. Topics covered: Biomedical imaging, Imaging devices; Light microscopy, Confocal microscopy; Regression and parameter estimation, Image processing and time series analysis (MATLAB); Hands on sessions with fluorescence microscopy.

## Workshop Attended

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3. *International Conference on Systems Biology*, IIT Bombay, Workshop sessions on **Noise and information transmission in Biological System, Stochastic processes in biology, an introduction, Multiscale modeling with MOOSE**.
4. *Midstate's Consortium for Mathematics and Science*, New faculty workshop, Hope College, Michigan, Holland, Jul. 2011.
5. *Teaching Center Introduction to STEM pedagogies workshop*, "Designing and facilitating Inquiry-Based Laboratory Teaching" Washington University in St Louis, 2012.

## Supervision of Students (B Tech Hons/M tech)

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Graduated 8 M. Tech students in Chemical Engineering Department  
Graduated 9 M Tech Students in Polymers and Biosystems Engineering  
Graduated 6 PhD Students in Chemical Engineering  
Currently supervising 6 PhD students

M Tech students (higher studies): Washinton University, St Louis, National University of Singapore, Singapore.  
M tech student placement: Production Engineer, Indian oil Corporation Limited ; AI Engineer, Akaike Technology, Japan;  
Data Engineer, Quantum. Data Engineer, Daftcraft, Inc: Scientist, Reddy's lab, AI/ML Engineer, Mialo.

## B tech Mentoring

|    |                       | <b>B tech Honors Project/Mini project</b>                         | <b>Higher studies</b>                                                                   | <b>Current position</b>                                                            |
|----|-----------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 1. | Jophin George Joseph  | Microfluidic device design for multiple cell types                | PhD, University of Michigan, Ann Arbor, Mechanobiology                                  | Senior Scientist, Zoetis Inc, medicine and vaccination, Michigan.                  |
| 2. | Apoorva Katragadda    | Mathematical Modeling of cell-to-cell communication               | PhD, National University of Singapore                                                   | Co-Founder, EmerStat, developed a medical device to reduce hemorrhage , Singapore. |
| 3. | Ravutla Suryateja     | Calcium signaling in hippocampal neurons                          | PhD, Georgia Tech university                                                            | Ongoing                                                                            |
| 4. | Kevin George          | Parameter estimation using dynamic optimization                   | German Centre for Artificial Intelligence (DFKI), Biologically informed neural networks | Ongoing                                                                            |
| 5. | DevAjith V S          | Image classification using Res-Net (CNN)                          | Master's at Cardiff University, Wales, UK                                               | CERN, The European organization for nuclear research, Switzerland                  |
| 6. | Mohammed Adil Ibrahim | Design of microfluidic Chip for monitoring cell to communications | PhD, Cornell University Precision medicine and genomics.                                | Ongoing                                                                            |
| 7. | Vikash Upadhyay       | Parameter estimation for biochemical oscillations                 | PhD, Pennsylvania State University                                                      | Ongoing                                                                            |

11. Infrastructure established at IITH (those with institute/JICA funds and those from individual project funds should be specified separately)

| S. No            | Description of infrastructure                                                                                            | Quantity | Funding        | Price     | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------|----------|----------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Institute</b> |                                                                                                                          |          |                |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 1.               | Laser scanning Confocal Microscope, Leica SP8. (Live cell imaging facility)<br><br>Training program on confocal imaging. | 01       | Institute Fund | 300 lakhs | Proposal writing for the live cell imaging facility, planning, procurement and establishment of the facility.<br>This facility was used by different departments including biomedical engineering, Biotechnology, Chemistry, Electrical Engineering and Materials Engineering department from 2014 to 2022. Significant amount of revenue was generated through this.<br>Trained 8 graduate students (PhD, M Tech) from my lab who have been performing confocal imaging of the samples from other departments. |

| <b>JICA (Japan International Co-operation Agency)</b> |                                                                                                             |       |           |           |                                                                                                                                                                                                                    |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------|-----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.                                                    | Multimode Palte reader and imaging facility<br>(Biological characterization facility)                       | 01    | JICA fund | 200 lakh  | Purchase Committee member                                                                                                                                                                                          |
| <b>National Facility</b>                              |                                                                                                             |       |           |           |                                                                                                                                                                                                                    |
| 1.                                                    | SATHI Microscopy Center, IIT Hyderabad<br>(Nikon- AXR Confocal with NSPARC)                                 | 01    | DST SATHI | 415 lakhs | Contributed to the SATHI proposal writing for application of real time fluorescence microscopy in life science.<br>Also. I was a part of the expert committee for purchase of Super resolution optical microscopy. |
| <b>Project</b>                                        |                                                                                                             |       |           |           |                                                                                                                                                                                                                    |
| 1.                                                    | Biological analysis facility: Spectrophotometer and Nanodrop (Estimation of concentration in small volume)  | 01+01 | Project   | 30 lakh   | Planning the purchase and setting up the facility                                                                                                                                                                  |
| 2.                                                    | Imaging facility: Fluorescent Microscopy (Nikon, Eclipse, Ti2 Main), Digital Sight 50 M Microscope Camera). |       |           | 50 lakh   |                                                                                                                                                                                                                    |
| 3.                                                    | Tissue culture Facility (BSL2) Perfusion culture with Syringe pump                                          | 02    | Project   | 35 lakh   | Planning the purchase and setting the facility                                                                                                                                                                     |
| 4.                                                    | Fermenter Facility (Project)                                                                                | 01    | Project   | 12 lakhs  | Planning the purchase and setting the facility                                                                                                                                                                     |
| 5.                                                    | Bioprinting Facility (Very recent)                                                                          | 01    | Project   | 11 lakhs  | Planning the purchase and setting the facility                                                                                                                                                                     |

## Research experience

- Confocal imaging of cytosolic calcium imaging in hippocampal neurons, monitoring of phase transitions in neurons, internalization of nanoparticles in cancer cells
- Non-animal models, Induction of hypoxia, development of chronic neurodegeneration model, development of Stroke model
- Modeling and simulation of viral infection process
- Applied systems biology and advanced live cell imaging techniques to investigate the role of G-protein subunit translocation in regulation of calcium oscillation in Hela cells
- AI/ML for automation in data analysis: DBSCAN based clustering of calcium spiking and classification of GPCR drug mediated calcium spiking, Application of CNN for automation in cell segmentation and video processing, Development of zero label techniques for cell segmentation, light weight CNN for smart phone-based detection of protein in serum, YOLO+Unet for detection of stress granules in the cells.
- Fabrication of imaging devices (microfluidic and milifluidic) using 3-D printing for disease diagnosis and screening of therapeutics
- Applied optogenetics to design input function to program immune cell migration and neuron differentiation
- Applied genetic engineering of baculovirus to mitigate mutations responsible for reduction in viral and recombinant protein yield during passage in insect cell culture.
- Analyzed system-level properties and quantified key protein levels under diseased conditions (diabetes and obesity) in insulin signaling pathway through mathematical modeling and simulations.

## References

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